

Program 3:

Code a Dockerized Python Flask or Node.js application

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Project Directory:

```
prog3
|- requirements.txt
|- app.py
|- Dockerfile
```

Program code and steps for execution:

Step 1: Create a directory with the prog3 name in order to execute the program

```
> mkdir prog3
```

```
> cd prog3/
```

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~$ cd devops_lab/
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab$ mkdir prog3
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab$ cd prog3/
```

Step 2: Now create the file called

1. requirements.txt 2. app.py 3. Dockerfile

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog3$ touch requirements.txt app.py Dockerfile
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog3$ ls
app.py  Dockerfile  requirements.txt
```

Step 3: All the file content will be as followed below

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog3$ nano requirements.txt
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog3$ cat requirements.txt
Flask==2.2.5
```

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog3$ nano app.py
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog3$ cat app.py
from flask import Flask
app = Flask(__name__)
```

```
@app.route("/")
def hello():
    return "Hello from Flask in Docker!"
```

```
if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000, debug=False)
```

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog3$ nano Dockerfile
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog3$ cat Dockerfile
# Use a small, stable Python base
FROM python:3.11-slim

# Set working dir
WORKDIR /app

# Copy requirements and install
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

# Copy application code
COPY app.py .

# Expose port and run
EXPOSE 5000
CMD ["python", "app.py"]
```

Step 4: Build and run the image and container respectively using the command,

> `docker build -t prog3 .`

> `docker run -d -p 5000:5000 prog3`

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog3$ docker build -t prog3 .
[+] Building 8.3s (11/11) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 334B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim 7.7s
=> [auth] library/python:pull token for registry-1.docker.io       0.0s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                       0.0s
=> [internal] load build context                                   0.0s
=> => transferring context: 290B                                      0.0s
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:7cd0079a9bd8800c 0.0s
=> => resolve docker.io/library/python:3.11-slim@sha256:7cd0079a9bd8800c 0.0s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> CACHED [3/5] COPY requirements.txt .                             0.0s
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt 0.0s
=> CACHED [5/5] COPY app.py .                                       0.0s
=> exporting to image                                              0.4s
=> => exporting layers                                              0.0s
=> => exporting manifest sha256:dc5cf1801a77f783a2b3270c8d1d96d7ec7cef57 0.0s
=> => exporting config sha256:c08b1101a36a581afad162dfa5b09de795af029924 0.0s
=> => exporting attestation manifest sha256:ddfc2a3879286dc4585a525fe056 0.0s
=> => exporting manifest list sha256:86ebdb51bf4fcdf587f6c74488e0294b4e 0.0s
=> => naming to docker.io/library/prog3:latest                     0.0s
=> => unpacking to docker.io/library/prog3:latest                  0.3s
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog3$ docker run -d -p 5000:5000 prog3
186b6bf7a8bf2ca41d3a0af2d1d6ea7279c1685ca59e3863179508b34f4d5352
```

Step 5: Open any browser and search for ,

> <http://localhost:5000/>

The output must be as shown below

